

HEALING WOUNDS

ARTIFICIAL SKIN

Our skin is our largest organ, covering and protecting everything inside our bodies. It is our first line of defense against injury or infections. Small injuries to it can heal easily, but big ones, such as serious burns, are harder to heal. Sometimes healthy skin can be taken from another part of the body and placed on a wound to help it repair itself, but this is difficult to do when the wound is severe. One solution is using artificial skin, which scientists can grow in a lab.

TREATING CUTS

For small wounds, an alternative to a bandage is a hydrocolloid dressing. These are made of substances that swell and form a gel when in contact with a wound. They are flexible and can be used for areas such as elbows, where typical bandages don't stick well.



Skin tissue

The piece of artificial skin below has been grown from cells donated by a patient. The cells are put in a gel, which supplies them with nutrients that allow them to multiply. The strip that forms is made up of a large number of cells and is now described as skin tissue. It can be applied to injuries using stitches or surgical glue.



THE SKIN

Our skin protects us from infection and helps regulate our body temperature, and nerves in the skin help us sense our surroundings using touch. Its thin, outermost layer is called the epidermis, and below it is the dermis, which contains blood vessels and nerves. Underneath these is a layer of fat, which acts as an energy store.

